



Macéo NARBONNET

Final-Year Engineering Internship — Software Engineering

6 months, starting February/March 2027

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Education

- Master's Degree in Computer Engineering (Engineering Degree) — Embedded & Autonomous Systems** 2024 – 2027
Université de Technologie de Compiègne (UTC) Compiègne, France
- Erasmus Exchange Semester** Sept. 2026 – Jan. 2027
Mälardalen University (MDU) Västerås, Sweden
- Preparatory Classes for Engineering Schools (CPGE TSI) — Maths, Physics & Engineering** 2022 – 2024
Lycée Gaspard Monge Chambéry, France
- French Baccalauréat (High School Diploma) — Technological Track (STI2D)** 2022
Lycée Gaspard Monge / Louis Armand Chambéry, France

Experience

- R&D Intern — AI-Based Motion Capture and 3D Modeling** Sept. 2025 – Feb. 2026
BMBI Laboratory (CNRS/UTC) — BioMov-E Team Compiègne, France
 - Improved a markerless motion capture pipeline: camera calibration, AI-based 2D keypoint detection, triangulation.
 - Characterized system errors against a Vicon reference system: mean standard deviation of 15.6 mm.
 - Integrated filtering, LSTM-based marker augmentation and OpenSim biomechanical correction: RMSE $7.6^\circ \rightarrow 5.2^\circ$.
 - Worked in a pair (Git/GitLab) and coordinated a team of students for multi-system test campaigns.
- IT Access & Identity Management (IAM) Assistant** Jul. – Aug. 2025
Crédit Agricole — Cross-Functional Tools & Security Unit, Transformation, Innovation & Quality Department Chambéry, France
 - Processed access and authorization (IAM) tickets for employees across the entire regional bank.
 - Improved the tracking of access-right changes, strengthening traceability and the unit's quality of service.
- Bank Branch Customer Service Assistant** Jul. – Aug. 2024
Crédit Agricole — Cognin Branch Chambéry, France

Projects

- Traffic Sign Detection** — Python, Scikit-learn, OpenCV, computer vision May – Jun. 2026
 - Multi-scale sliding-window detection pipeline with Random Forest classification (HOG + HSV + LBP features).
 - Multi-scale hard negative mining and dual-criterion NMS (IoU + containment): validation AP 0.095 $\rightarrow$ 0.293.
- Preemptive Real-Time Kernel — Priority Inversion** — C, real-time systems, ARM Cortex-M7, QEMU Apr. 2026
 - Preemptive scheduler with dynamic priorities in C (TCB redesign, linked queues) on ARM Cortex-M7 (QEMU).
 - Priority Inheritance Protocol mutex: blocking time bounded to a single critical section.
- Autonomous Mapping with a Mobile Robot** — Python, ROS 2, LiDAR, computer vision, robotics Mar. – Jun. 2025
 - ROS 2 nodes (Python) detecting directional arrows via HSV thresholding on the rectified camera stream.
 - Multi-sensor fusion (encoders, gyroscope, IMU, LiDAR) and scan-to-scan ICP for real-time odometry.
- 3D Trajectory Modeling by Stereo Vision** — Python, OpenCV, computer vision, image processing 2023 – 2024
 - Stereo calibration (intrinsic/extrinsic, distortion) and geometric triangulation; applied to video refereeing.

Skills

- Programming:** C, C++, Python, MATLAB, SQL, Bash
- Embedded & Real-Time Systems:** ARM Cortex-M7, bare-metal, scheduling, QEMU, GDB, embedded Linux
- Computer Vision & AI:** OpenCV, Scikit-learn, NumPy/SciPy, CNN, YOLO, CUDA
- Robotics:** ROS 2, LiDAR, IMU, encoders, gyroscope, multi-sensor fusion, ICP
- Software Engineering & Tools:** OOP, Design Patterns, UML, Qt, Git/GitLab, Linux, $\LaTeX$
- Languages:** French (native), English (C1), Italian (A2)

Extracurricular Activities

- Head of Sports** — UTC Student Sports Association (22 sports clubs, 150 competing athletes)
- Communications & Equipment Manager** — Student Audiovisual Association
- Tennis Lead & Team Captain** — UTC Tennis Club and Team
- WWOOF Volunteer** — Farm maintenance / livestock, Canada (Vancouver)